

PERSONAL STUDY PLAN · 2026–2028

AI Engineering Master Timetable

Target: AI Infra Engineer · DSA + Maths + GATE DA + 4-Phase Roadmap · ~22 Months

Phase 1 · Foundations · Month 1–6

Phase 2 · ML/DL · Month 7–12

Phase 3 · GenAI/LLMs · Month 13–17

Phase 4 · AI Infra · Month 18–22

Phase 1Phase 2Phase 3Phase 4DSAGATE DAMathematicsProjectsRevisionWind-down / Rest

PHASE	DURATION	CORE TOPICS	DSA TRACK	GATE DA	MILESTONE
Phase 1 — Foundations	Month 1–6	Python advanced, async, type hints · Linux/Bash · Git · FastAPI · PostgreSQL · Docker · Redis · Linear Algebra · Calculus · Stats	Arrays, Strings, Hash Maps, Two Pointers, Sliding Window, Linked Lists, Stacks, Queues, Binary Search, Trees, BST	60 min/weeknight · 3 hr Saturday mock · Discrete Math + Probability focus	Dockerised FastAPI service live. 150+ LeetCode solved.
Phase 2 — ML/DL	Month 7–12	Scikit-learn · NumPy/Pandas · PyTorch · CNN/RNN/Transformers from scratch · HuggingFace · MLflow · W&B · Kaggle	Heaps, Graphs (BFS/DFS/Dijkstra/Topo), DP (knapsack, LCS), Bit Manipulation, Tries, Segment Trees	Continue daily. ML/stats maps to Phase 2. Ramp before exam date.	Mini-GPT trained. BERT fine-tuned. Kaggle top 30%. 300+ solved.
Phase 3 — GenAI/LLMs	Month 13–17	Production RAG · Hybrid search · RAGAS evals · QLoRA fine-tuning · LangGraph agents · vLLM · Quantization · Inference optimization	Hard DP + Graph combos. System design 1×/week. Mock interviews bi-weekly.	Final 2 months before exam: 3 hr/day. After exam: replace with infra time.	RAG platform deployed. Fine-tuned Llama on vLLM. Agent eval harness built.
Phase 4 — AI Infra	Month 18–22	CUDA/Triton kernels · Flash Attention · FSDP/DeepSpeed · Megatron-LM · Kubernetes + Kubeflow · Triton Inference Server · Prometheus/Grafana	Concurrent DS, LRU/LFU, Hard interview problems. Mock interview weekly.	GATE done. Slots replaced with project/infra time.	Custom CUDA kernel. Distributed training run. K8s autoscaling live.

ACTUAL DAILY SCHEDULE (MON–THU)

- **5:00–6:00 am** Exercise
- **6:00–7:30 am** Prep + commute
- **7:30–4:30 pm** Office
- **4:30–5:30 pm** Commute home
- **5:30–7:00 pm** Wind-down — rest, freshen up, dinner. NO study.
- **7:00–11:30 pm** Study window = **4.5 hours**
- **11:30 pm** Hard bedtime. Non-negotiable.
- **Fri / Sat / Sun** No constraint → 8–9 hr study available

CORE PRINCIPLES

- Wind-down 5:30–7:00 pm is sacred — no guilt, no skipping
- Never study same subject >90 min straight
- 10-min break between every block
- **Sunday = Revision + Projects ONLY. No new content.**
- DSA is daily — even on hard days, minimum 1 problem
- Anki flashcards reviewed every Sunday
- GitHub push every week without fail
- Mon/Wed = roadmap heavy · Tue/Thu = DSA/Math heavy

GATE DA STRATEGY

- Phase 1–2 weeknights: 60 min (last block before bed)
- Friday: 3 hr full mock (free day, morning slot)
- Saturday: 2 hr mock/topic block
- Phase 3 final 2 months before exam: bump weeknight to 90 min
- After exam: all GATE slots → infra/project time
- **GATE overlaps roadmap:** Maths, Discrete Math, Probability, Linear Algebra, Algorithms — every session is a double win
- Primary resource: previous year papers. Build formula sheet weekly.

Mon–Thu: Study starts 7:00 pm (after wind-down). Window = 4.5 hrs → 3 blocks of 80–90 min with breaks. Mon/Wed = roadmap heavy. Tue/Thu = DSA/Math heavy. Fri–Sun = full day, 8–9 hrs available.

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
5:30 pm	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study. No guilt.	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study. No guilt.	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study. No guilt.	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study. No guilt.	Fri: Free day START 9:00 AM No commute. Full day study session.	Sat: Free day START 9:00 AM No commute. Full day study session.	Sun: Revision only START 9:00 AM No new content. Revision + projects only.
7:00 pm (9 am Fri–Sun)	7:00–8:30 pm PYTHON CORE Async, type hints, decorators, context managers. Roadmap module order.	7:00–8:30 pm DSA — THEORY FIRST VisuAlgo animation → Jenny/Abdul Bari video → 2 problems (1 easy + 1 medium)	7:00–8:30 pm FASTAPI + DOCKER FastAPI docs → REST design → Docker basics → Compose setup	7:00–8:30 pm DSA — PRACTICE 2 medium problems timed (35 min each). No hints. Verbal explanation after.	9:00 am–12:00 pm GATE DA — FULL BLOCK 3 hr focused. Discrete Math, Probability, CS fundamentals. Previous year papers only.	9:00–11:00 am DSA — EXTENDED 4 problems: 2 medium + 1 hard attempt. NeetCode review after each.	9:00–11:00 am DSA REVISION Anki flashcards. Re-solve week’s hardest from memory. No new content ever.
8:30 pm (11 am Fri–Sun)	8:30–8:45 pm BREAK Stand, stretch, water. Step away from screen.	8:30–8:45 pm BREAK Stand, stretch, water.	8:30–8:45 pm BREAK Stand, stretch, water.	8:30–8:45 pm BREAK Stand, stretch, water.	12:00–1:00 pm LUNCH + REST Full break. Walk if possible.	11:00 am–1:00 pm GATE DA — MOCK Full mock section or topic deep-dive. Discrete Math overlaps roadmap math.	11:00 am–1:00 pm PROJECT BUILD Phase 1 project: FastAPI + PostgreSQL + JWT + Docker. Push to GitHub.
8:45 pm (1 pm Fri–Sun)	8:45–10:00 pm DSA — PRACTICE 2 problems. LeetCode 75 plan order. Focus on week’s pattern. No hints 30 min.	8:45–10:00 pm MATHEMATICS 3Blue1Brown (Lin Alg / Calculus) → Khan Academy drills → MML book exercises	8:45–10:00 pm DSA — PRACTICE 2 problems. Explain solution aloud after solving. Add patterns to Anki.	8:45–10:00 pm MATHEMATICS Linear Algebra: matrix ops, eigenvalues. Stats: Bayes, distributions. Implement in NumPy.	1:00–3:00 pm BACKEND PROJECT FastAPI + Docker. Auth, error handling, pytest tests. Ship something every session.	1:00–2:00 pm LUNCH + REST Full break. No screens.	1:00–2:00 pm LUNCH + REST Full break.
10:00 pm (2 pm Fri–Sun)	10:00–10:10 pm BREAK Short break before final block.	10:00–10:10 pm BREAK Short break.	10:00–10:10 pm BREAK Short break.	10:00–10:10 pm BREAK Short break.	3:00–4:30 pm DSA — 3 PROBLEMS Review previous mistakes. 3 problems timed. NeetCode review after each.	2:00–4:30 pm PHASE 1 FLAGSHIP PROJECT FastAPI + PostgreSQL + Redis + Docker Compose + pytest. Deploy to VPS.	2:00–4:00 pm MATH REVISION Implement gradient descent, PCA from scratch. Re-read hard sections. No new topics.
10:10 pm (4:30 pm Fri–Sun)	10:10–11:30 pm GATE DA 60 min: prev year questions. Discrete Math + Probability. Build formula sheet.	10:10–11:30 pm LINUX / GIT / BASH MIT Missing Semester · Bash scripts · OverTheWire Bandit challenges	10:10–11:30 pm MATHEMATICS 3Blue1Brown video → implement concept in code → Brilliant.org problems	10:10–11:30 pm GATE DA 60 min: Linear Algebra + Stats — overlaps roadmap math. Double win.	4:30–6:00 pm MATH — IMPLEMENT Code matrix multiply, PCA, gradient descent from scratch. No libraries.	4:30–6:00 pm MATH — IMPLEMENT SVD, eigendecomposition, probability problems on Brilliant.org. Visualise with matplotlib.	4:00–6:00 pm PROJECT POLISH Clean code, write README, add tests, push GitHub. Document learnings this week.
					6:00–7:00 pm REST Mandatory. No screens.	6:00–7:00 pm REST Walk, dinner, no screens.	6:00–7:00 pm WEEKLY LOG 3 things learned. 3 to redo. Plan next week. GitHub streak check.
					7:00–9:00 pm LINUX DEEP DIVE Bash scripts: log analyzer, system monitor, file processor. OverTheWire.	7:00–8:30 pm GATE DA EVENING Discrete Math + Probability overlap. Formula sheet review and additions.	

Wind-down 5:30–7:00 pm still protected on weekdays. Study still starts 7:00 pm = 4.5 hrs. DSA continues daily — now harder (graphs, DP). Math deepens to backprop calculus. GATE DA continues. Phase 2 replaces Python/backend slot with ML/DL theory and PyTorch.

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
5:30 pm	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study.	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study.	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study.	5:30–7:00 pm WIND-DOWN Rest, freshen up, dinner. No study.	Free day START 9:00 AM Full day available.	Free day START 9:00 AM Full day available.	Revision only START 9:00 AM No new content.
7:00 pm (9 am Fri–Sun)	7:00–8:30 pm ML THEORY StatQuest videos → scikit-learn code-along → implement algorithm from scratch	7:00–8:30 pm DSA — GRAPHS + DP Abdul Bari theory → 2 medium/hard. BFS/DFS/ Dijkstra/DP patterns.	7:00–8:30 pm PYTORCH Karpathy Zero to Hero — code along every video. Build neural net from scratch.	7:00–8:30 pm DSA — HARD PROBLEMS 2 hard problems timed (40 min each). System design warm-up 10 min.	9:00 am–12:00 pm GATE DA — FULL MOCK 3 hr full mock or topic block. ML/stats maps to Phase 2. Two birds, one stone.	9:00–11:00 am DSA — MOCK STYLE 3 medium + 1 hard. Timed strictly. Verbal explanation after each.	9:00–11:00 am DSA REVISION Anki review. Re-solve 5 hardest from week from memory.
8:30 pm (11 am Fri–Sun)	8:30–8:45 pm BREAK Stretch, water.	8:30–8:45 pm BREAK Stretch, water.	8:30–8:45 pm BREAK Stretch, water.	8:30–8:45 pm BREAK Stretch, water.	12:00–1:00 pm LUNCH + REST Full break.	11:00 am–1:00 pm PAPER READING "Attention is All You Need" → implement multi-head attention in PyTorch line by line.	11:00 am–1:00 pm KAGGLE COMPETITION Active competition. Feature engineering focus. Document approach in notebook.
8:45 pm (1 pm Fri–Sun)	8:45–10:00 pm DSA PRACTICE 2 medium/hard. Week's topic. No hints first 30 min. Explain aloud after.	8:45–10:00 pm DL PROJECT BERT fine-tuning / image classifier / mini-GPT. Code every single session.	8:45–10:00 pm DSA PRACTICE 2 problems. Add new patterns to Anki. Verbal explanation always.	8:45–10:00 pm HUGGINGFACE / MLFLOW HF NLP course chapters → MLflow experiment tracking → W&B sweep setup	1:00–3:30 pm PYTORCH PROJECT Train on GPU (Colab/Kaggle). Log with W&B. Deploy as FastAPI + Docker.	1:00–2:00 pm LUNCH + REST Full break. No screens.	1:00–2:00 pm LUNCH + REST Full break.
10:00 pm (2 pm Fri–Sun)	10:00–10:10 pm BREAK Short break.	10:00–10:10 pm BREAK Short break.	10:00–10:10 pm BREAK Short break.	10:00–10:10 pm BREAK Short break.	3:30–5:00 pm DSA — 3 PROBLEMS Mock interview style. Verbal explanation. Strict timer.	2:00–5:00 pm PHASE 2 FLAGSHIP Mini-GPT (nanoGPT). Train on GPU. Understand every line of attention + training loop.	2:00–4:00 pm ML/DL REVISION Revisit: attention math, backprop, bias-variance. Re-read hard sections only.
10:10 pm (4:30 pm Fri–Sun)	10:10–11:30 pm GATE DA 60 min: ML/stats portion. GATE sections that map directly to Phase 2 syllabus.	10:10–11:30 pm MATHEMATICS Backprop calculus: chain rule, Jacobians. Implement backprop from scratch in NumPy.	10:10–11:30 pm GATE DA 60 min: algorithms + data structures — maps directly to DSA track. Prev year papers.	10:10–11:30 pm MATHEMATICS Probability: MLE, MAP, distributions. Info theory: entropy, KL divergence.	5:00–6:00 pm REST / WALK Mandatory. No screens. 6:00–8:30 pm HUGGINGFACE NLP COURSE Tokenizers, fine-tuning, model hub. Deploy on HF Spaces with Gradio.	5:00–6:30 pm GATE DA MOCK Full section mock. Review all mistakes. Build error log. 7:00–8:30 pm MATH — IMPLEMENT PCA from eigenvectors. MLE for Gaussian. Visualise with matplotlib.	4:00–6:00 pm PROJECT — DEPLOY Deploy model as FastAPI + Docker. Add validation, logging, test endpoints. 6:00–7:00 pm WEEKLY LOG 3 learnings. 3 to redo. GitHub streak. Plan next week.

Weekday study window still 7:00–11:30 pm = 4.5 hrs. GATE DA: ramp to 90 min/night in final 2 months before exam. After exam → replace with project time. DSA = hard level only. System design 1×/week.

Day / Time	Block 1 (90 min)	Block 2 (90 min)	Block 3 (60 min)
Mon & Wed 5:30–7 pm: Wind-down Study: 7:00–11:30 pm	RAG / Agents / Fine-tuning LlamaIndex, LangGraph, Unsloth QLoRA. Code every session.	DSA — 2 Hard Problems 40 min each. No hints. Verbal explanation required.	Math + LLM Project Info theory / optimization → 30 min project progress
Tue & Thu 5:30–7 pm: Wind-down Study: 7:00–11:30 pm	DSA + System Design 2 hard problems + 1 system design warm-up (15 min)	vLLM / Inference / Agents PagedAttention, LangGraph multi-agent, eval harness	GATE DA 60 min weekdays. 90 min in final exam crunch 2 months.
Friday 9 am – 9 pm (free day)	GATE DA Full Mock (3 hr) Full mock test 9 am–12 pm. Review immediately after.	Flagship RAG Sprint Hybrid search, streaming, Redis cache, production deploy	DSA Mock Interview 2 problems, 45 min each, zero hints, strict timer. Verbal only.
Saturday 9 am – 9 pm (free day)	DSA — 3 Hard + Sys Design Timed hard problems + 1 full system design (45 min)	Fine-tuning Project QLoRA + Unsloth. Eval vs base model with LLM-as-judge.	GATE DA Revision Weak areas from Friday mock. Formula cheat sheet additions.
Sunday Revision + Projects only	DSA Revision (2 hr) → Portfolio: blog post, GitHub README, deploy demo		Revise LLM/RAG concepts. Weekly log: 3 learnings, plan next week.

GATE done — those slots now go to infra projects. Weekday window still 7:00–11:30 pm. DSA = hard interview level. System design weekly. Mock interview bi-weekly. Every project must show GPU or distributed expertise.

Day / Time	Block 1 (90 min)	Block 2 (90 min)	Block 3 (60 min)
Mon & Wed 5:30–7 pm: Wind-down Study: 7:00–11:30 pm	CUDA / Triton / GPU PMPP book + CUDA kernels. Matrix multiply, reduction, flash attention kernels.	DSA — 2 Hard Problems Interview simulation. Timed. Explain aloud. No partial credit mindset.	Distributed Training DeepSpeed ZeRO, FSDP, Megatron-LM pipeline parallelism
Tue & Thu 5:30–7 pm: Wind-down Study: 7:00–11:30 pm	System Design Interview 1 full system: distributed LLM serving / training infra. Design + critique.	K8s / Kubeflow / Triton Deploy on K8s. Autoscaling. Triton Inference Server setup.	Observability Prometheus + Grafana + OpenTelemetry. Cost attribution. Alerts.
Friday 9 am – 9 pm (free day)	Infra Project Sprint (3 hr) Custom CUDA kernel or distributed training run. Deep work only.	Full Mock Interview 2 hard DSA + 1 system design. Record yourself. Watch back and review.	CUDA Profiling Flash Attention impl. Profile with nsight. Optimise memory access patterns.
Saturday 9 am – 9 pm (free day)	DSA — Hard + Codeforces 3 hard LeetCode + 1 Codeforces Div2 C/D problem. Timed.	Deep Project Inference optimizer / custom kernel / K8s full stack. Ship something.	Paper Reading Flash Attention, FSDP, Megatron, Efficient GPU Memory. Derive key results by hand.
Sunday Revision + Projects only	DSA Revision (2 hr) — hardest 10 re-solved from memory → Full system design re-done from scratch		Flagship project polish + case study + GitHub + personal website update

Period	Topics	LeetCode Focus	Primary Resources	Daily Target
Month 1–2 Phase 1 start	Arrays, Strings, Hash Maps, Two Pointers, Sliding Window, Linked Lists, Stacks, Queues	LeetCode 75 plan. Easy → Medium. 1 easy + 1 medium/day.	VisuAlgo, Jenny's Lectures, Apna College, NeetCode YouTube, Grokking Algorithms book	2 problems/day weekday. 3–4 Saturday. Anki every Sunday.
Month 3–4 Phase 1 mid	Binary Search, Sorting, Trees, BST, Recursion basics, Backtracking intro	Top Interview 150. 2 mediums/day. 1 hard attempt per week.	NeetCode YouTube, Abdul Bari, GeeksForGeeks theory, VisuAlgo trees	2 problems/day. Saturday: 4 problems timed mock style.
Month 5–6 Phase 1 end	Heaps, Priority Queue, Graphs (BFS/DFS/Topo), Union-Find, Recursion + Backtracking	Graph + Backtracking sections. 1 medium + 1 hard/day.	Abdul Bari graphs, NeetCode, CP Algorithms, Neso Academy	2 problems/day. Saturday: 3 medium + 1 hard. Friday: review week mistakes.
Month 7–9 Phase 2	DP (knapsack, LCS, matrix chain, interval), Dijkstra, Bellman-Ford, Bit Manipulation, Tries	DP Blind 75. Advanced graph problems. 2 hard/day target.	Abdul Bari DP, Aditya Verma DP YouTube, CP Algorithms, Striver SDE Sheet	2 hard/day. Saturday: 3 hard + 1 system design warm-up.
Month 10–12 Phase 2 end	Segment Trees, Fenwick Trees, Hard DP (digit DP, bitmask), Advanced graphs, Monotonic Stack	Hard LeetCode + Codeforces Div 2 C/D. Mock interviews weekly.	CP Algorithms, Codeforces, Striver A-Z sheet, competitive contests	2 hard/day. Bi-weekly mock interview: 2 problems + verbal. No hints ever.
Month 13–22 Phase 3 + 4	Concurrent DS, LRU/LFU caches, Graph+DP combos, Interview patterns (AI Infra lens)	Company lists (Google, NVIDIA, Meta, OpenAI). Mock interviews weekly.	Blind 75, NeetCode 150, company Glassdoor, Pramp mock interviews	2 hard/day. Weekly full mock: 2 hard + 1 system design, timed, recorded.

DSA 3-STEP WORKFLOW (EVERY PROBLEM)

→ **Step 1 — Understand (10–15 min):** VisuAlgo animation → GFG article → Jenny/Abdul Bari video. Know WHAT and WHY before touching code.

→ **Step 2 — Attempt (20–30 min):** Brute force first, then optimise. Code it. Test it. No solution lookup — 30 min hard cap.

→ **Step 3 — Review (10 min):** NeetCode YouTube. Compare approaches. If yours was worse, understand why. Add pattern to Anki card.

→ **Never skip Step 1.** Most stagnation = jumping to problems without understanding the concept first.

→ **Sunday Anki rule:** Every new pattern becomes a card. Reviewed every Sunday without exception.

PHASE COMPLETION CHECKLISTS

→ **Phase 1 done:** Write async Python with full type hints. Dockerised FastAPI + PostgreSQL live. Complex SQL. Confident in Linux/Bash.

→ **Phase 2 done:** Build transformer from scratch. Fine-tune BERT. Track experiments with W&B. Explain attention mechanism from memory.

→ **Phase 3 done:** Production RAG with RAGAS evals. Fine-tuned Llama on vLLM. Multi-agent system. 3× latency improvement documented.

→ **Phase 4 done:** Profile GPU workloads with nsight. Distributed training run working. K8s with autoscaling live. Observability stack. Custom CUDA kernel written.

RESOURCE QUICK REFERENCE

→ **Math:** 3Blue1Brown → Khan Academy → MML book (ch 2–6) → StatQuest

→ **DL:** Karpathy Zero to Hero → fast.ai → d2l.ai → Goodfellow (reference)

→ **LLMs:** HF NLP course → LlamaIndex docs → vLLM docs → LoRA/QLoRA papers

→ **Infra:** PMPP book (CUDA) → DeepSpeed docs → NVIDIA dev blog → K8s official

→ **DSA:** VisuAlgo + Jenny + Abdul Bari (theory) → NeetCode (solution review)

→ **GATE DA:** Previous year papers (primary) → NPTEL → Made Easy material

→ **System Design:** DDIA book + Gaurav Sen YouTube + High Scalability blog

→ **GPU papers:** Flash Attention, PagedAttention, LoRA, FSDP, Megatron-LM